COREPRESENTABILITY OF THE ISOMORPHISM FUNCTOR BY A CLOSED IMMERSION IN THE FINITE PRESENTATION FLAT AND PURE CASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The situation and functors for a map of modules entering the flatterling statement. Same chapter flat.tex, situation labelled situation-iso (statement, proof line 6673). Fixes the map u : F -> G and the functor F_iso whose representability by a closed immersion Z -> S is asserted.
- Universal purity from purity along a fibre and the sheaf and limit preserving properties of the functors. Same chapter flat.tex, labels lemma-finite-type-flat-pure-along-fibre-is-universal (proof line 6689), lemma-iso-sheaf (proof lines 6691 and 6747) and lemma-iso-sheaf part (4) (proof line 6747); Descent, labels descent-lemma-closed-immersion (proof line 6693) and descent-lemma-descent-data-sheaves (proof line 6694); Limits, remark limits-remark-limit-preserving (proof line 6749). Gives that F is universally pure so the hypotheses may be used after any base change, lets the question be treated locally for the etale topology through the sheaf property and the descent of closed immersions, and supplies the limit preserving property that makes the representing object of finite presentation.
- The affine etale neighbourhood with projective global sections, the purity criterion and constructibility of the relative assassin. Same chapter flat.tex, labels lemma-finite-presentation-flat-along-fibre (proof line 6701), lemma-affine-locally-projective-pure (proof line 6714), lemma-criterion (proof line 6716) and lemma-universally-separating (proof line 6735); More on Morphisms, label more-morphisms-lemma-relative-assassin-constructible (proof

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

line 6725); Morphisms, label morphisms-lemma-constructible-containing-open (proof line 6728); Same chapter flat.tex, label lemma-flattening-module-map (proof line 6742). Produces the affine etale neighbourhood and the etale g with projective global sections, places the relative associated points inside the image of g so that they can be removed after shrinking, reduces injectivity to the pulled back map, and finally identifies the representing ideal with the one constructed by the module map flatterling lemma.

Source reference. Original lemma 03I0: sources/stacks-chow/tags.txt, source lines 4586–4586. Navigation pointer for \ref{descent-lemma-closed-immersion} used inside the counted span (line 6693); the argument that consumes it is retained in the span.

Source reference. Original lemma 02W5: sources/stacks-chow/tags.txt, source lines 3792–3792. Navigation pointer for \ref{descent-lemma-descent-data-sheaves} used inside the counted span (line 6694); the argument that consumes it is retained in the span.

Source reference. Original lemma 05K6: sources/stacks-flat/pinned-original-source.tex, source lines 5148–5148. Navigation pointer for \ref{lemma-affine-locally-projective-pure} used inside the counted span (line 6714); the argument that consumes it is retained in the span.

Source reference. Original lemma 05L4: sources/stacks-flat/pinned-original-source.tex, source lines 5197–5197. Navigation pointer for \ref{lemma-criterion} used inside the counted span (line 6716); the argument that consumes it is retained in the span.

Source reference. Original lemma 05KW: sources/stacks-flat/pinned-original-source.tex, source lines 3342–3342. Navigation pointer for \ref{lemma-finite-presentation-flat-along-fibre} used inside the counted span (line 6701); the argument that consumes it is retained in the span.

Source reference. Original lemma 05L5: sources/stacks-flat/pinned-original-source.tex, source lines 5276–5276. Navigation pointer for \ref{lemma-finite-type-flat-pure-along-fibre-is-universal} used inside the counted span (line 6689); the argument that consumes it is retained in the span.

Source reference. Original lemma 05PE: sources/stacks-flat/pinned-original-source.tex, source lines 6646–6646. Navigation pointer for \ref{lemma-flattening-module-map} used inside the counted span (line 6742); the argument that consumes it is retained in the span.

Source reference. Original lemma 05MJ: sources/stacks-flat/pinned-original-source.tex, source lines 5919–5919. Navigation pointer for \ref{lemma-iso-sheaf} used inside the counted span (line 6691); the argument that consumes it is retained in the span.

Source reference. Original lemma 05PD: sources/stacks-flat/pinned-original-source.tex, source lines 6598–6598. Navigation pointer for \ref{lemma-universally-separating} used inside the counted span (line 6735); the argument that consumes it is retained in the span.

Source reference. Original lemma 05LX: sources/stacks-chow/tags.txt, source lines 7251–7251. Navigation pointer for \ref{limits-remark-limit-preserving} used inside the counted span (line 6749); the argument that consumes it is retained in the span.

Source reference. Original lemma 05KR: sources/stacks-chow/tags.txt, source lines 7210–7210. Navigation pointer for \ref{more-morphisms-lemma-relative-assassin-constructible} used inside the counted span (line 6725); the argument that consumes it is retained in the span.

Source reference. Original lemma 05LW: sources/stacks-chow/tags.txt, source lines 7250–7250. Navigation pointer for \ref{morphisms-lemma-constructible-containing-open} used inside the counted span (line 6728); the argument that consumes it is retained in the span.

Source reference. Original lemma 05MH: sources/stacks-flat/pinned-original-source.tex, source lines 5896–5896. Navigation pointer for \ref{situation-iso} used inside the counted span (line 6673); the argument that consumes it is retained in the span.

Theorem 0.1. *In Situation 05MH assume*

(1) *f is of finite presentation,*
(2) *$\mathcal{F}$ is of finite presentation, flat over S, and pure relative to S, and*
(3) *u is surjective.*

Then F_{iso} is representable by a closed immersion $Z \to S$. Moreover $Z \to S$ is of finite presentation if $\mathcal{G}$ is of finite presentation.

Proof. We will use without further mention that $\mathcal{F}$ is universally pure over S, see Lemma 05L5. By Lemma 05MJ and Descent, Lemmas 03I0 and 02W5 the question is local for the étale topology on S. Hence it suffices to prove, given $s \in S$, that there exists an étale neighbourhood of (S, s) so that the theorem holds.

Using Lemma 05KW and after replacing S by an elementary étale neighbourhood of s we may assume there exists a commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

of schemes of finite presentation over S, where g is étale, $X_s \subset g(X')$, the schemes X' and S are affine, $\Gamma(X', g^*\mathcal{F})$ a projective $\Gamma(S, \mathcal{O}_S)$-module. Note that $g^*\mathcal{F}$ is universally pure over S, see Lemma 05K6. Hence by Lemma 05L4 we see that the open $g(X')$ contains the points of $\mathrm{Ass}_{X/S}(\mathcal{F})$ lying over $\mathrm{Spec}(\mathcal{O}_{S,s})$. Set

$$E = \{t \in S \mid \mathrm{Ass}_{X_t}(\mathcal{F}_t) \subset g(X')\}.$$

By More on Morphisms, Lemma 05KR E is a constructible subset of S. We have seen that $\mathrm{Spec}(\mathcal{O}_{S,s}) \subset E$. By Morphisms, Lemma 05LW we see that E contains an open neighbourhood of s. Hence after replacing S by a smaller affine neighbourhood of s we may assume that $\mathrm{Ass}_{X/S}(\mathcal{F}) \subset g(X')$.

Since we have assumed that u is surjective we have $F_{iso} = F_{inj}$. From Lemma 05PD it follows that $u : \mathcal{F} \to \mathcal{G}$ is injective if and only if $g^*u : g^*\mathcal{F} \to g^*\mathcal{G}$ is injective, and the same remains true after any base change. Hence we have reduced to the case where, in addition to the assumptions in the theorem, $X \to S$ is a morphism of affine schemes and $\Gamma(X, \mathcal{F})$ is a projective $\Gamma(S, \mathcal{O}_S)$-module. This case follows immediately from Lemma 05PE.

To see that Z is of finite presentation if $\mathcal{G}$ is of finite presentation, combine Lemma 05MJ part (4) with Limits, Remark 05LX. □